Quiz 1: Practice Solutions

Lectures 1–3

Try the practice handout in 15 minutes before reading these solutions. Score each of its five parts as 2 practice points for a correct final answer and 0 otherwise.

1(a)	1(b)	1(c)	2(a)	2(b)
2	6	3	B	C

1. State and action values

6 points

In (a)–(b), π chooses `run`; $\gamma = \frac{1}{2}$ in all three parts.

(a) The worn machine

Running in W pays 1 and returns to W with probability 1:

$$V^\pi(\mathrm{W}) = 1 + \tfrac{1}{2}V^\pi(\mathrm{W}) \quad \Longrightarrow \quad \boxed{V^\pi(\mathrm{W}) = 2}.$$

W has a self-loop under this policy, but it pays a reward every step.

(b) The good machine

Average the two possible successor values in the Bellman equation:

$$\begin{aligned} V^\pi(\mathrm{G}) &= 4 + \tfrac{1}{2}\left(\tfrac{1}{2}V^\pi(\mathrm{G}) + \tfrac{1}{2}V^\pi(\mathrm{W})\right) \\ &= 4 + \tfrac{1}{4}V^\pi(\mathrm{G}) + \tfrac{1}{2}. \end{aligned}$$

Thus $\frac{3}{4}V^\pi(\mathrm{G}) = \frac{9}{2}$, giving $\boxed{V^\pi(\mathrm{G}) = 6}$.

(c) Independent MDP: take a once, then follow π

Action a pays -2 and leads to Y. From Y onward, the expected return under π is the supplied value 10:

$$Q^\pi(\mathrm{X}, a) = -2 + \tfrac{1}{2}V^\pi(\mathrm{Y}) = -2 + \tfrac{1}{2} \cdot 10 = \boxed{3}.$$

The immediate reward is counted once and is not discounted. The value of Y already includes all future rewards under π. No answer from (a) or (b) is used.

2. Concepts

4 points

(a) B: Contextual bandit

The visitor's attributes are the context, the ad is the action, and feedback is available only for the chosen ad. Ad choices do not affect future contexts or responses. No known click model or correct-action labels are supplied.

(b) C: Inventory alone is not Markov

The two histories give the same proposed state and action but different next-state distributions. This violates the Markov property. The state must distinguish the pending shipments. Making the ordering policy stochastic does not supply the missing shipment information.